

Cybersecurity Task-1:

Foundation & Environment Setup (Theory Part)

👉 Objective:

To build a basic understanding of cybersecurity concepts, networking, cryptography, and to set up a safe virtual lab for learning and testing security tools.

1. Fundamentals of Cybersecurity:

Cybersecurity is the practice of protecting systems, networks, and data from digital attacks, theft, or damage.

It ensures that information remains:

Confidential → only authorized people can access it.

Accurate (Integrity) → information is not changed by unauthorized people.

Available → systems and data are accessible when needed.

These three form the CIA Triad – the foundation of cybersecurity.

2. The CIA Triad:

Element	Meaning	Example
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👉 Confidentiality	Keeping information secret	Using passwords or encryption to protect files
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👉 Integrity	Ensuring data is not modified	Checking file hash values (MD5/SHA)
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👉 Availability	Making sure systems are always accessible	Preventing server crashes or DDoS attacks
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3. Common Cyber Threats:

Type	Description	Example
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👉 Phishing Tricking people into sharing sensitive info through fake emails or websites
A fake bank email asking for your password

👉 Malware Malicious software that harms computers Viruses, Trojans, Spyware

👉 Ransomware Locks your files until you pay a ransom WannaCry attack

👉 SQL Injection Inserting malicious SQL queries into web apps Stealing login data from a website

👉 DDoS Attack Flooding a website with traffic to crash it Botnet attack on servers

👉 Brute Force Attack Guessing passwords repeatedly Trying all combinations until login succeeds

4. Types of Hackers:

Type	Role	Example
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👉 White Hat	Ethical hackers who test systems legally	Security testers, ethical hackers
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👉 Black Hat	Illegal hackers with malicious intent	Hackers who steal data or money
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👉 Gray Hat	Hackers who explore systems without permission but without evil intent Report vulnerabilities without authorization	
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5. Networking Basics:

👉 Cybersecurity depends on understanding how data travels over a network.

👉 OSI Model Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

👉 IP Addressing: Identifies devices on a network (e.g., 192.168.1.1).

👉 DNS: Converts domain names into IP addresses.

👉 TCP/IP: Core protocol for communication over the internet.

6. Cryptography Basics:

👉 Cryptography is the science of protecting data using mathematical techniques.

Concept	Description
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👉 Encryption	Converting readable data into unreadable form
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👉 Decryption	Converting it back to readable form
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👉 Symmetric Encryption	Same key for encryption and decryption (e.g., AES)
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👉 Asymmetric Encryption	Different keys – Public and Private (e.g., RSA)
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👉 Hashing	Converts data to a fixed-length value; cannot be reversed (e.g., MD5, SHA-256)
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7. Important Cybersecurity Tools:

Tool	Purpose
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Wireshark :	Captures and analyzes network traffic
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Nmap :	Scans networks and finds open ports
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Burp Suite :	Tests web application vulnerabilities
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Metasploit:	Framework for penetration testing
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Netcat :	Debugs and tests network connections
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8. Outcome of Task-1

After completing this task, I will Understand the basics of cybersecurity and networking.

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SessionActionsEditViewHelp

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unix	3	[]	STREAM	CONNECTED	9979	/run/dbus/system_bus_socket
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unix	3	[]	STREAM	CONNECTED	9443	/run/user/1000/at-spi/bus_0

kali@kali: ~

Snipping Tool

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Session Actions Edit View Help

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Snipping Tool

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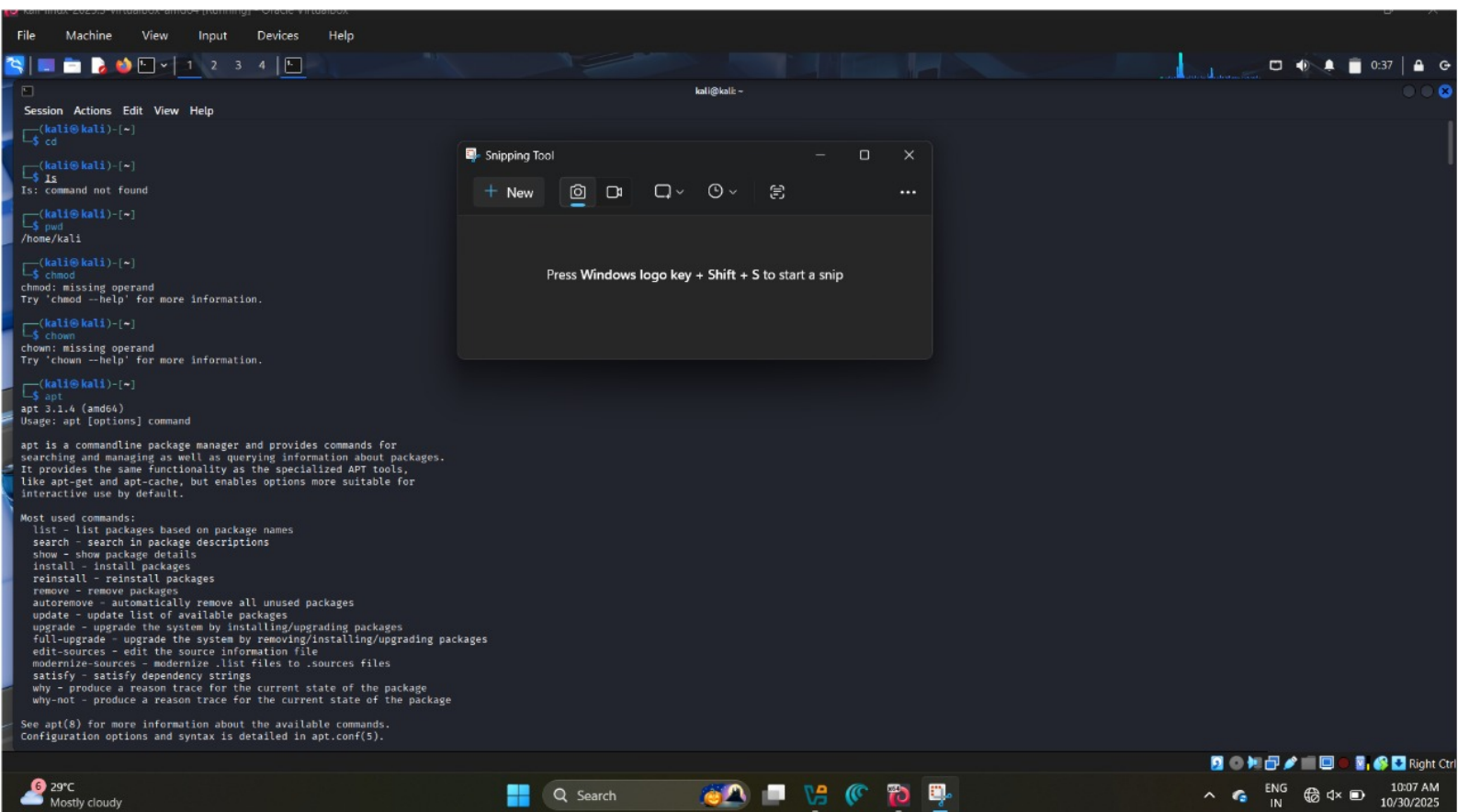
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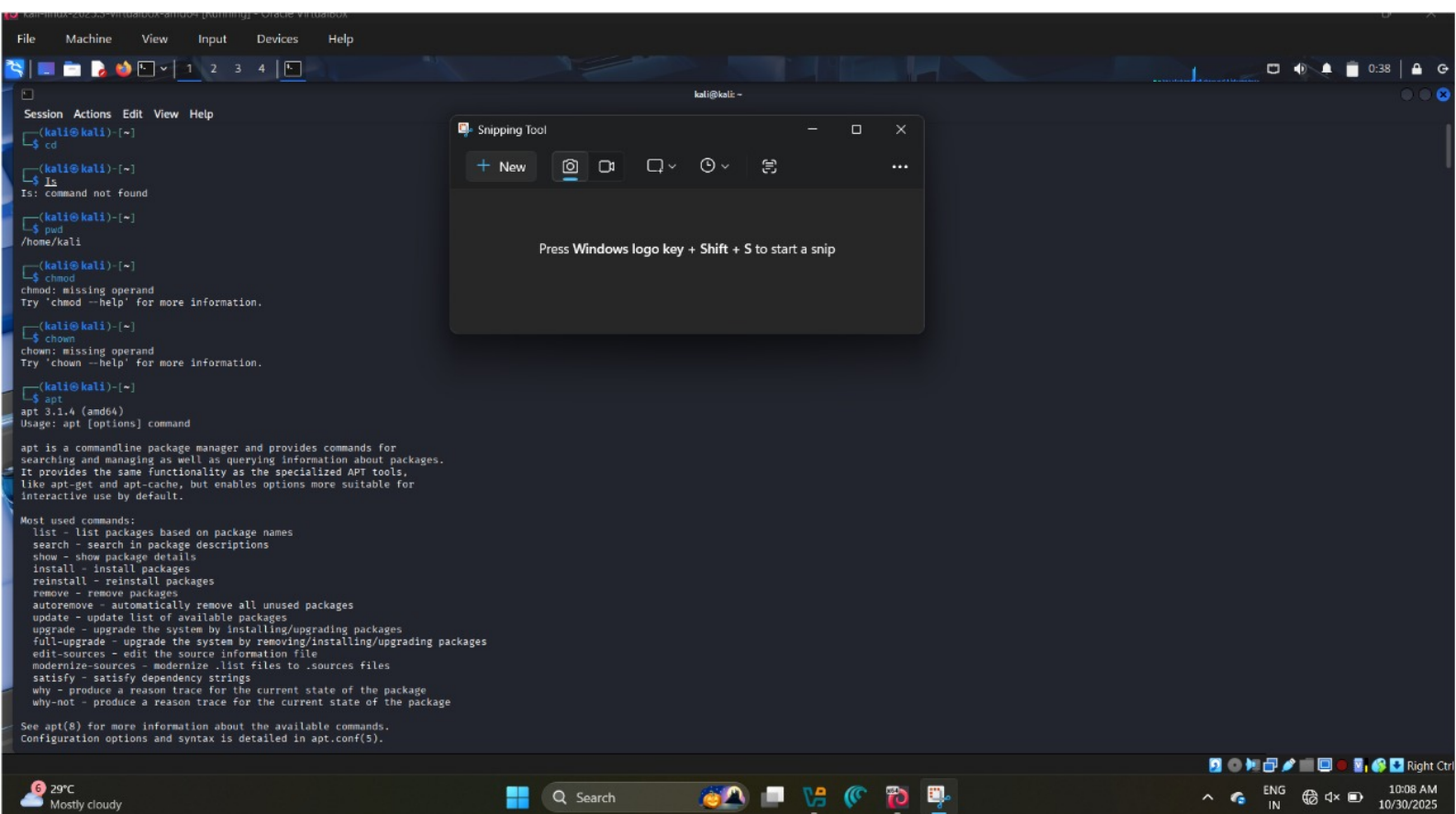
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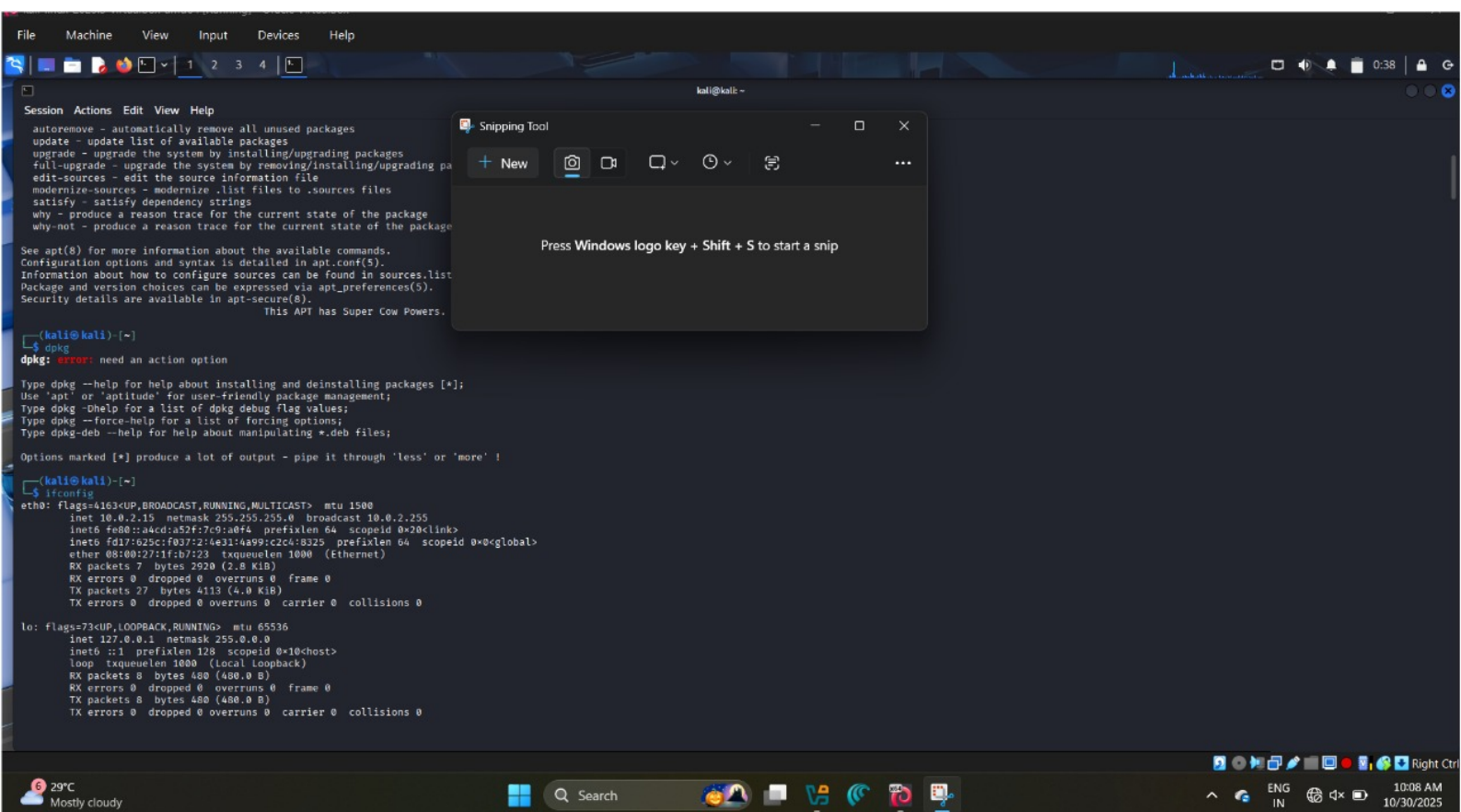
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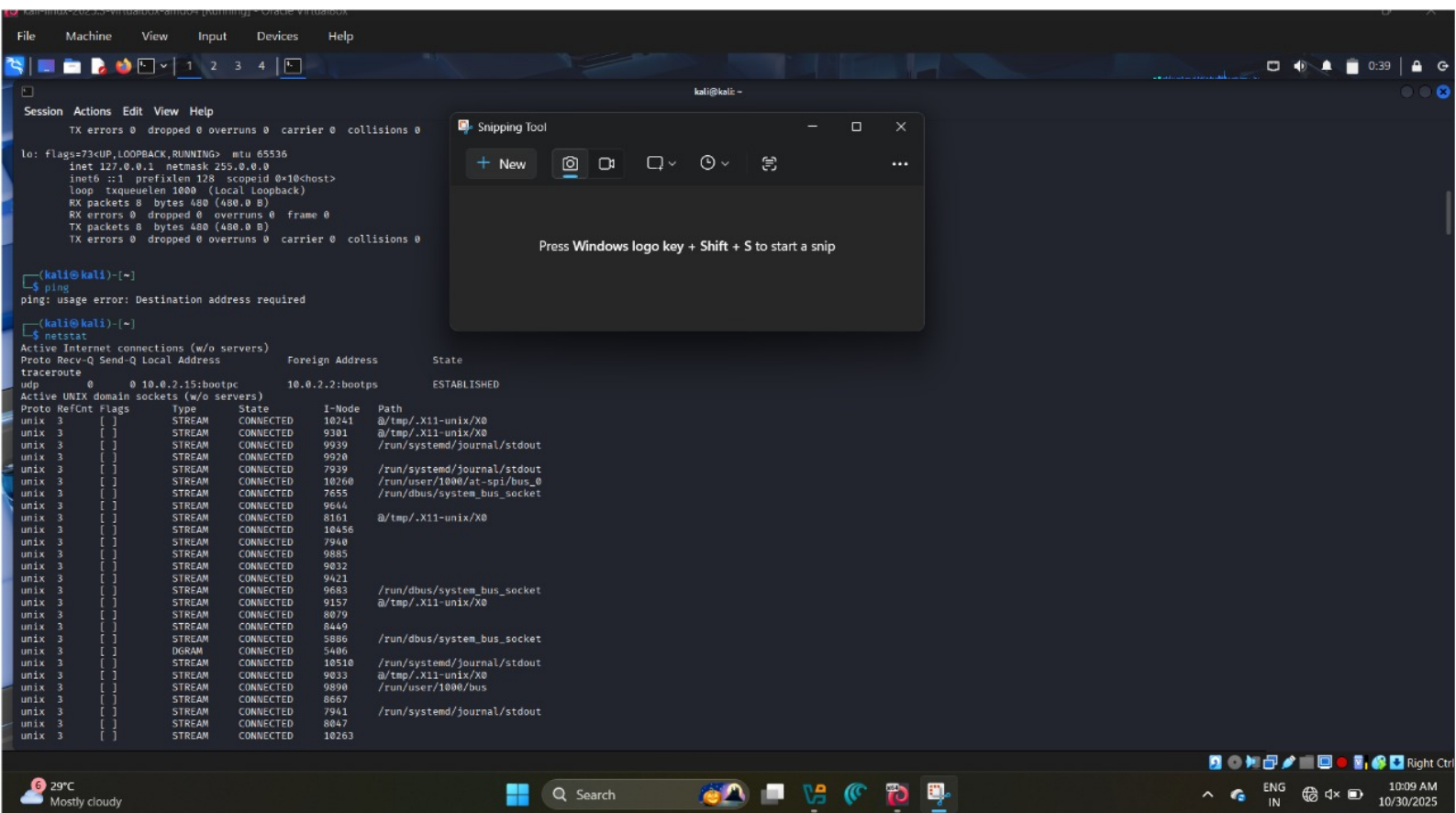
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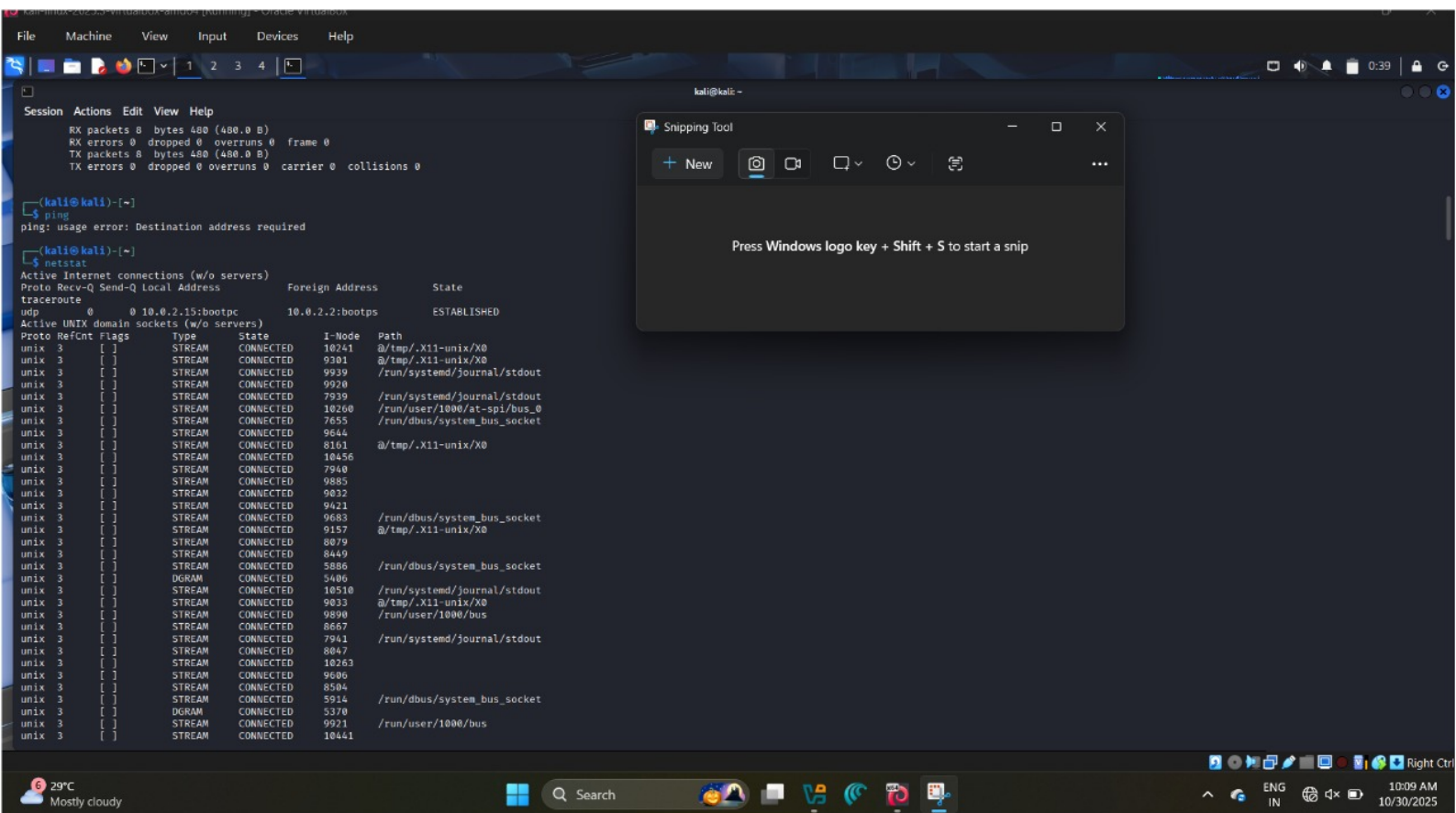
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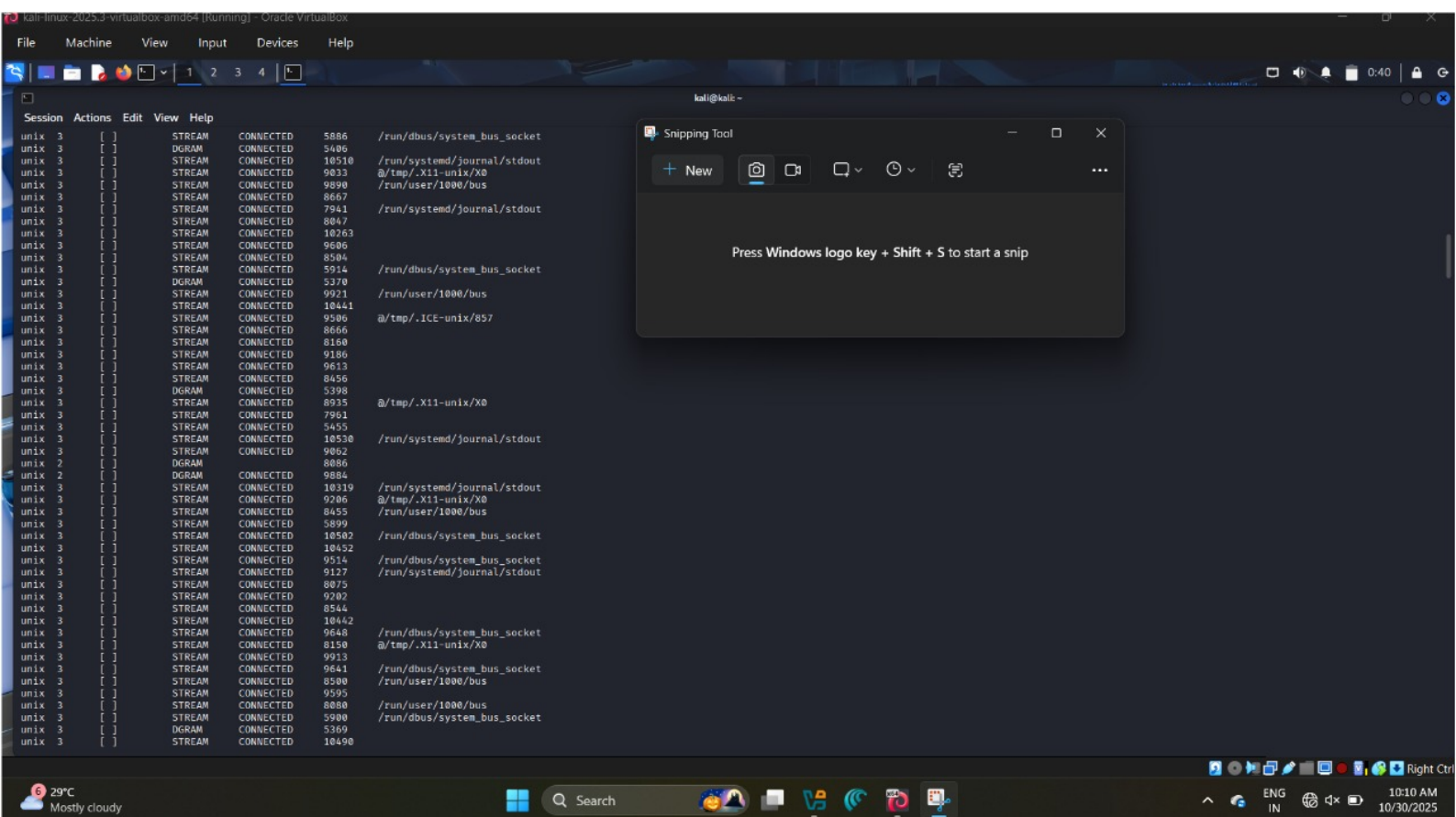


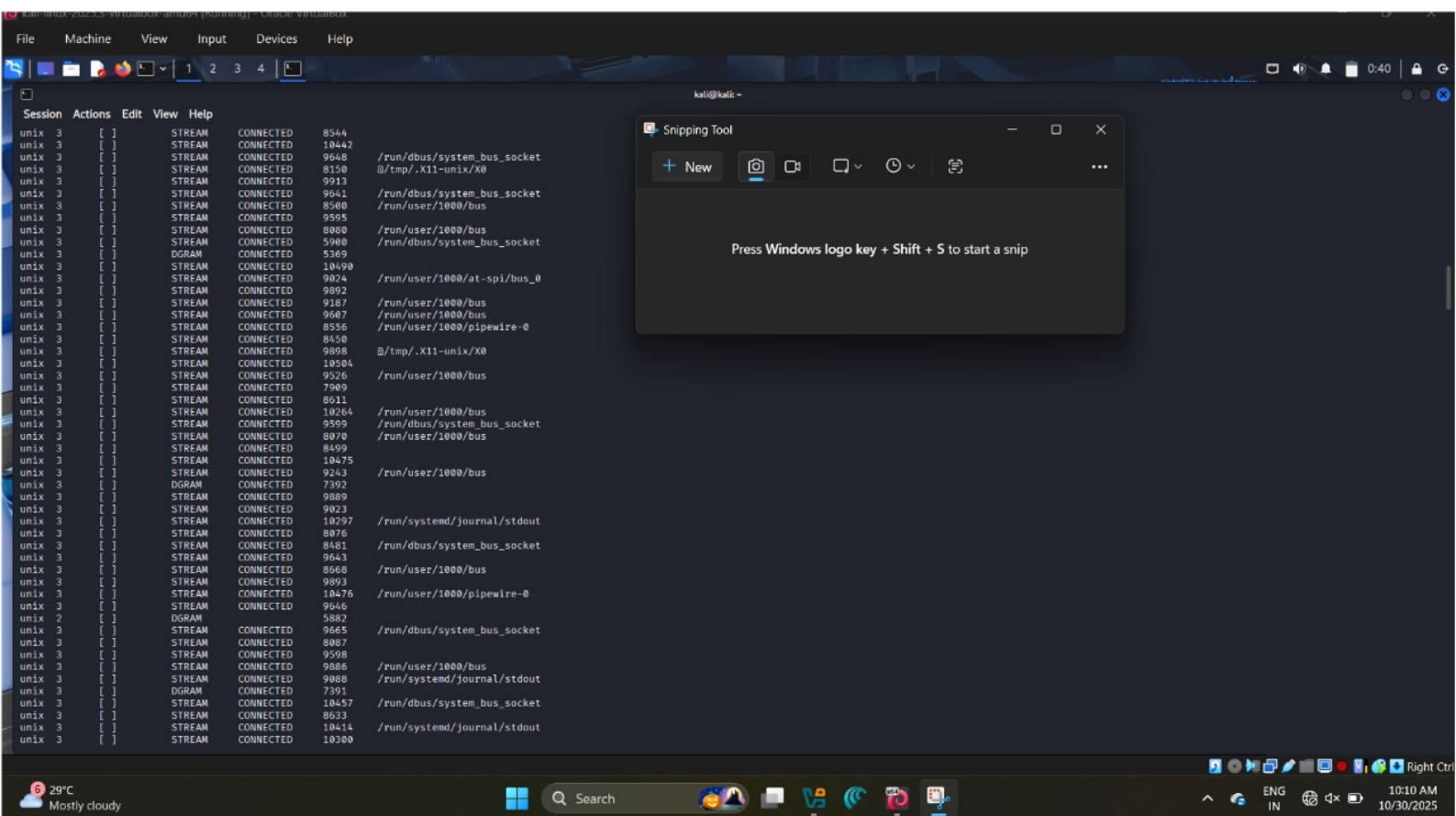












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SessionActionsEditViewHelp

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unix	3	[]	STREAM	CONNECTED	8502	/run/user/1000/bus
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unix	3	[]	STREAM	CONNECTED	7967	/run/systemd/journal/stdout
unix	3	[]	STREAM	CONNECTED	9941	/run/user/1000/bus
unix	3	[]	STREAM	CONNECTED	10473	/run/user/1000/pulse/native
unix	3	[]	STREAM	CONNECTED	10424	/run/dbus/system_bus_socket
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unix	3	[]	STREAM	CONNECTED	9435	/run/user/1000/at-spi/bus_0
unix	3	[]	STREAM	CONNECTED	8332	/run/dbus/system_bus_socket
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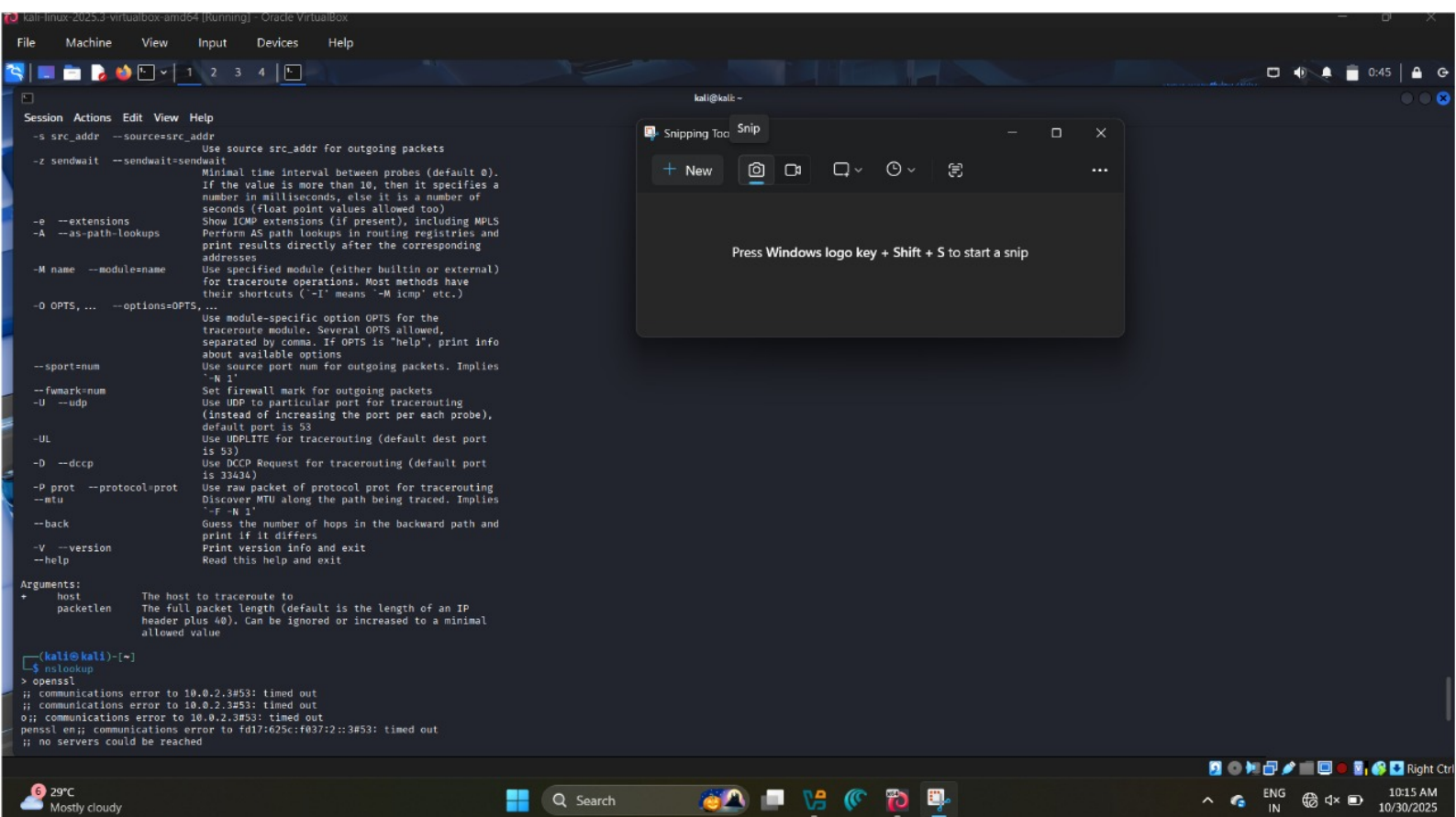
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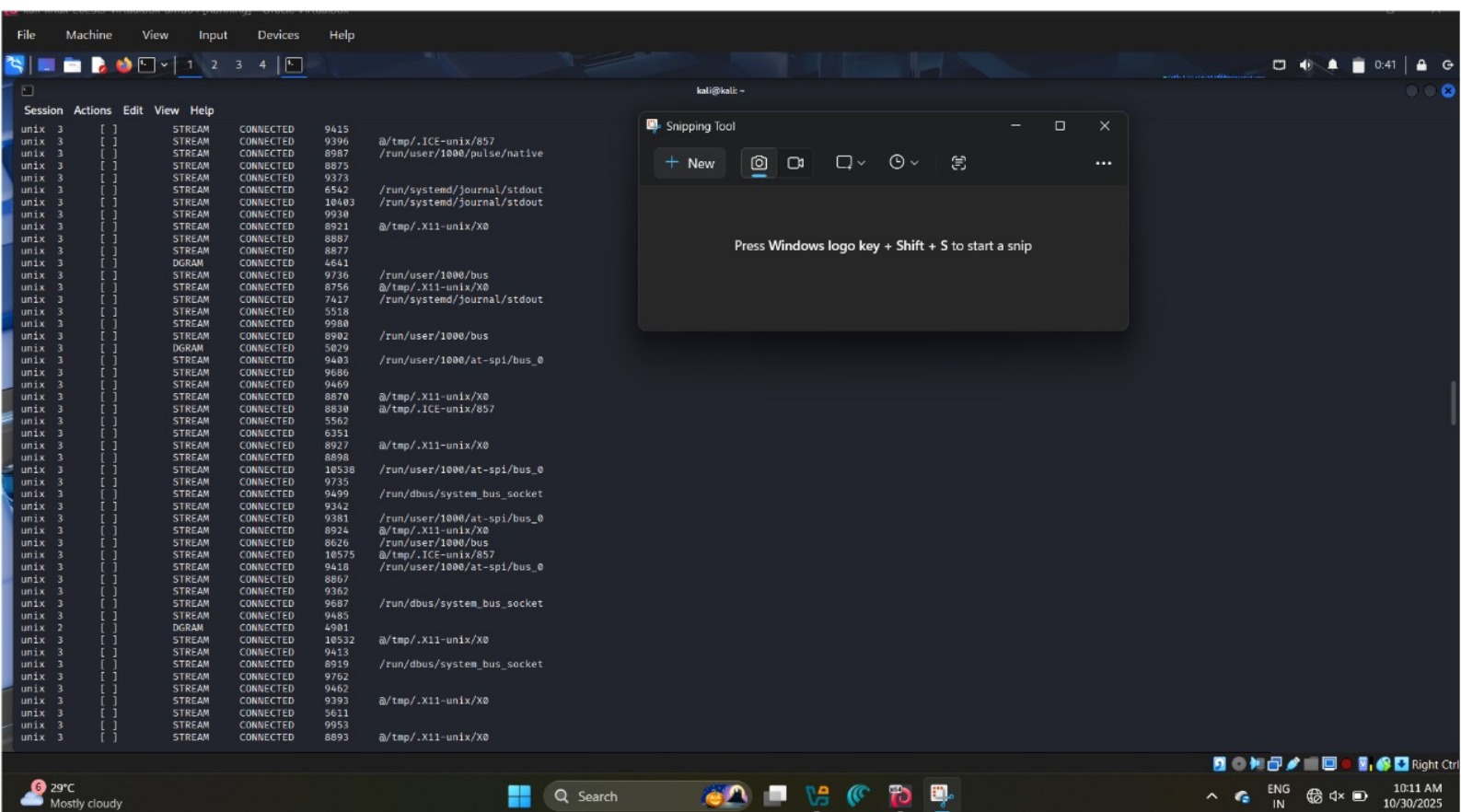
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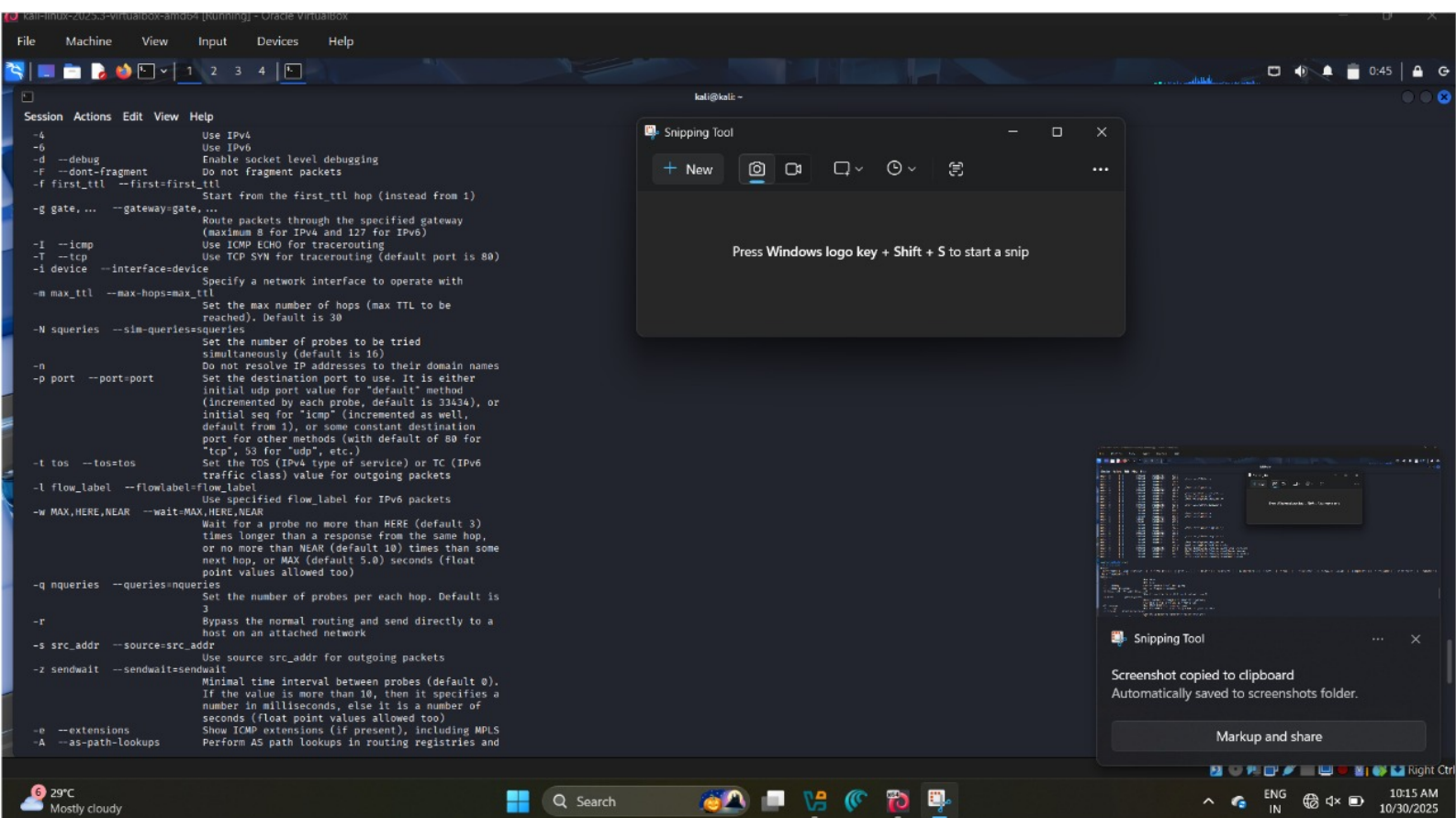
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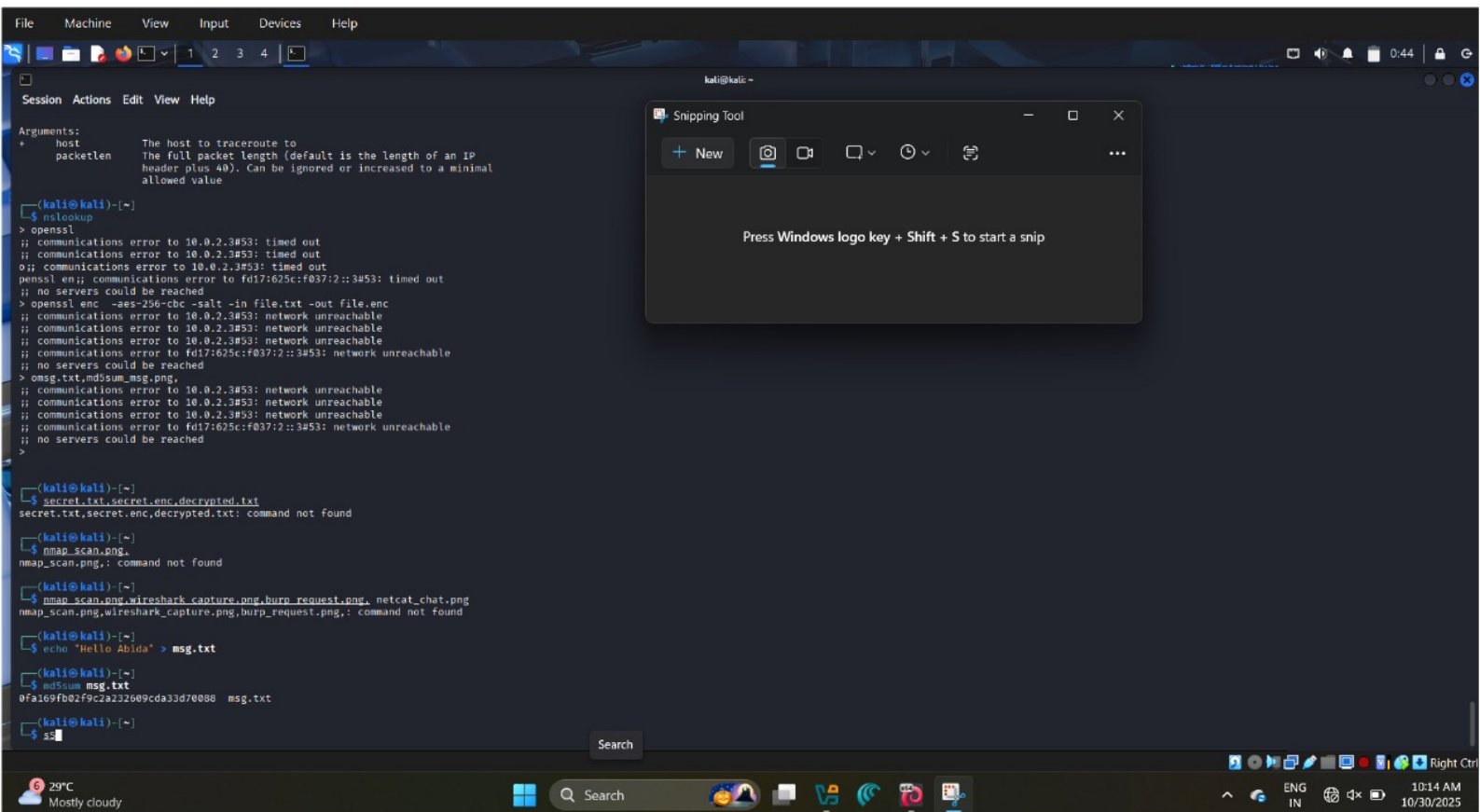
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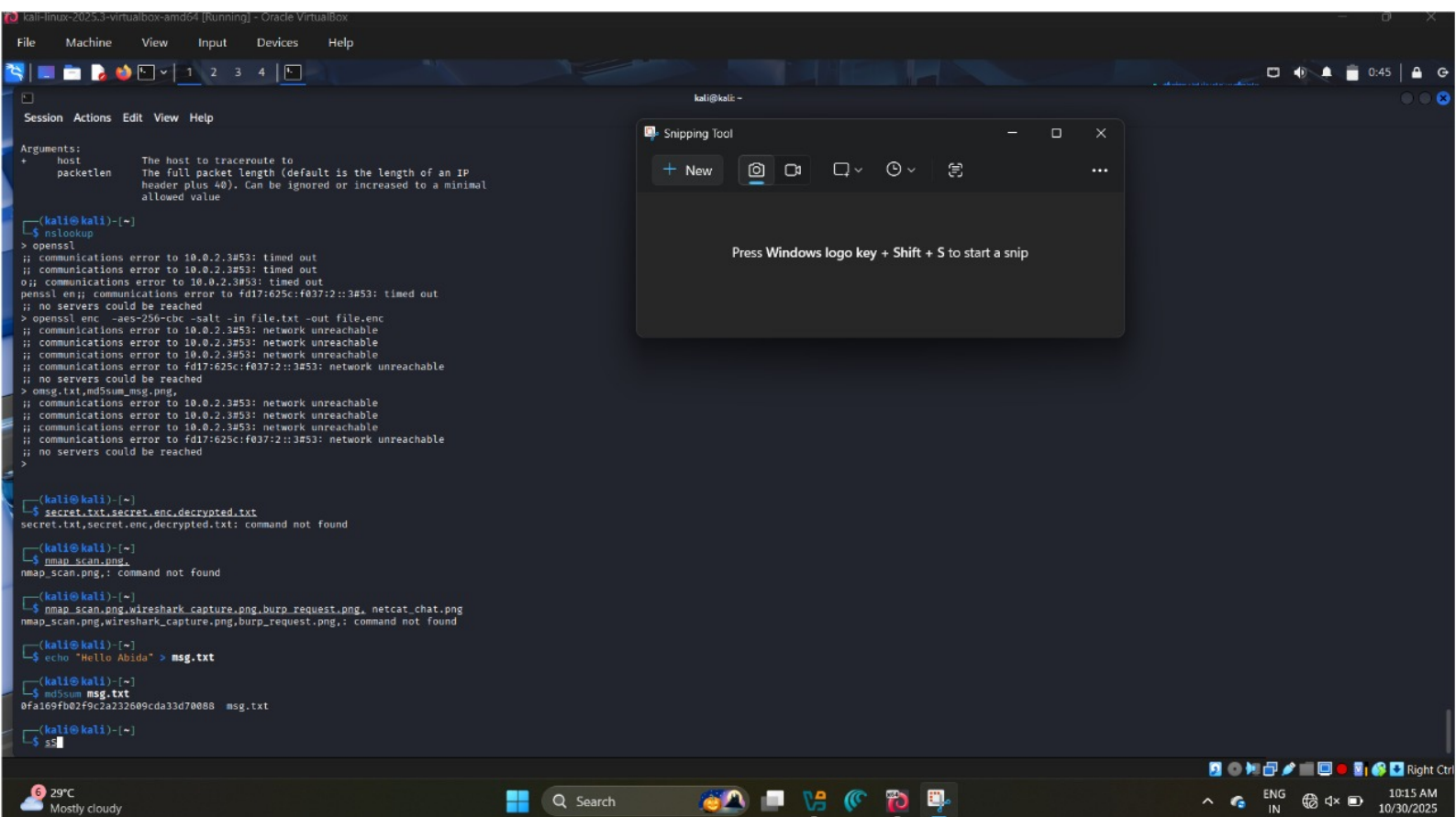
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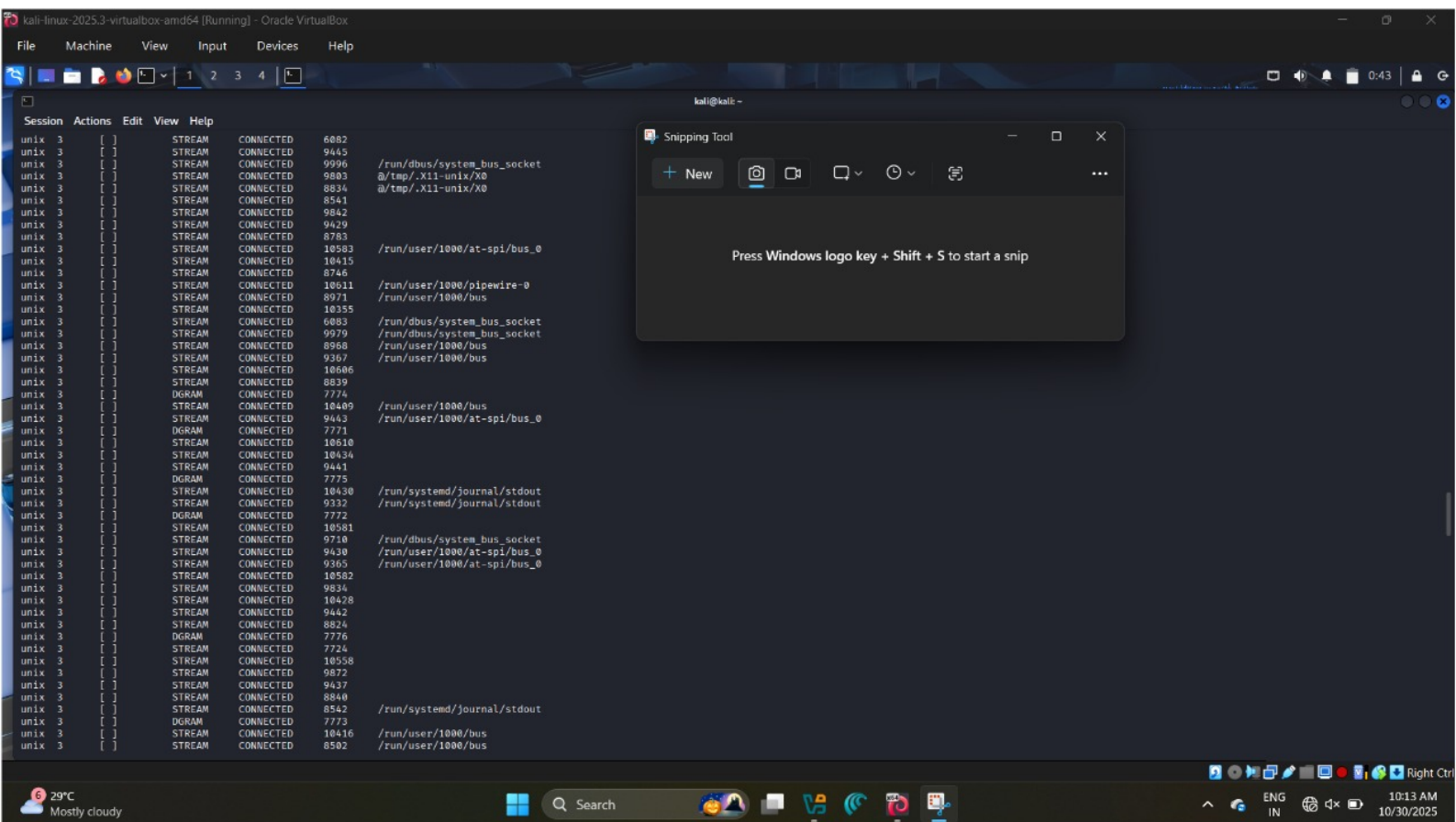


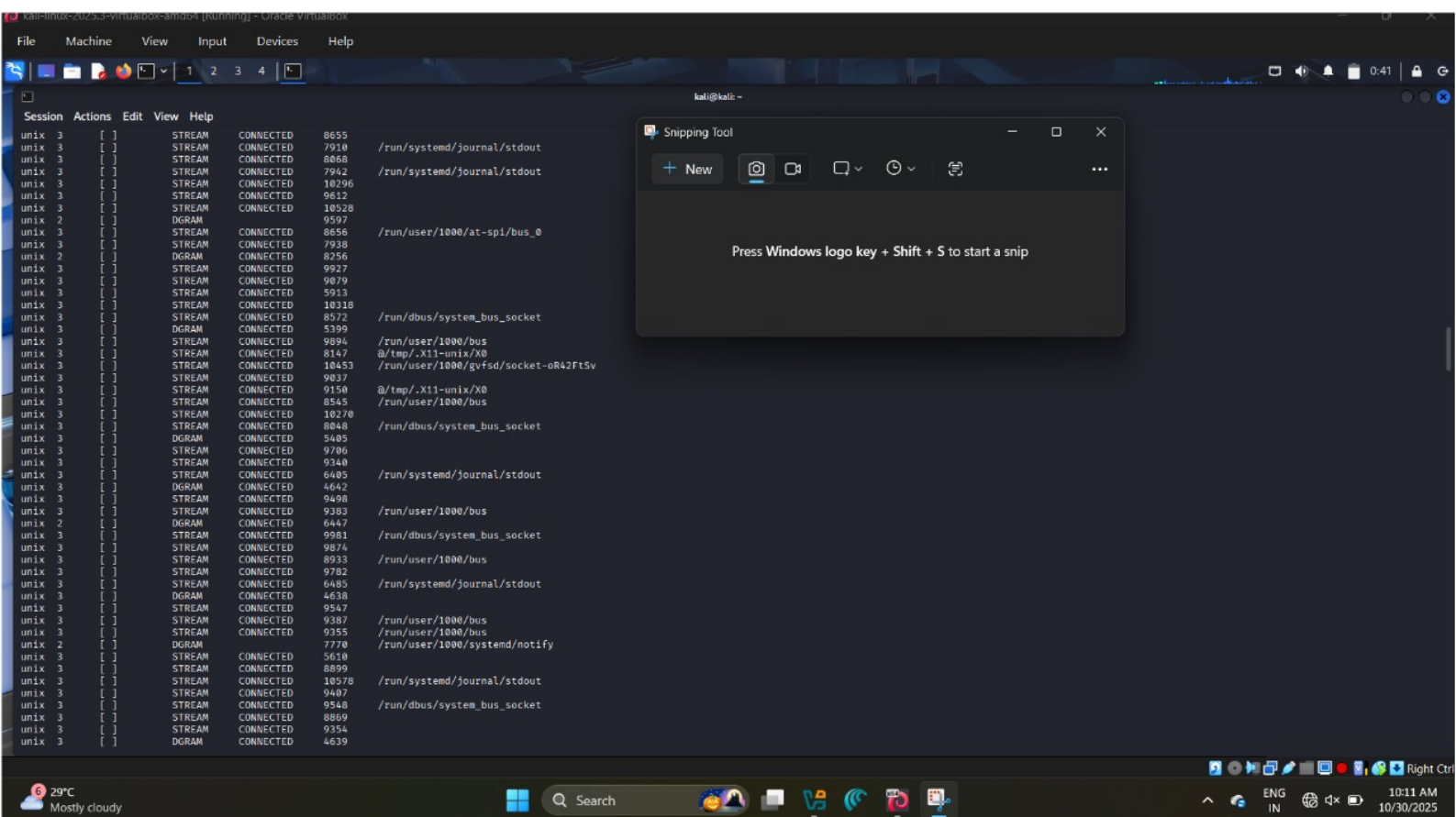












FileMachineViewInputDevicesHelp

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unix 3 [] STREAM CONNECTED 5611 @/tmp/.X11-unix/X0

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unix 3 [] STREAM CONNECTED 8893 @/tmp/.X11-unix/X0

unix 3 [] STREAM CONNECTED 8933 @/tmp/.X11-unix/X0

unix 3 [] STREAM CONNECTED 8549 /run/systemd/notify

unix 3 [] DGRAM CONNECTED 4636 /run/systemd/journal/stdout

unix 3 [] STREAM CONNECTED 9802 /run/user/1000/at-spi/bus_0

unix 3 [] STREAM CONNECTED 8876 /run/user/1000/bus

unix 3 [] STREAM CONNECTED 10261 /run/user/1000/bus

unix 3 [] STREAM CONNECTED 9235 /run/user/1000/bus

unix 3 [] STREAM CONNECTED 8931 /run/user/1000/bus

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unix 3 [] STREAM CONNECTED 7858 /run/user/1000/bus

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unix 3 [] STREAM CONNECTED 8912 /run/user/1000/at-spi/bus_0

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unix 3 [] STREAM CONNECTED 9400 /run/user/1000/bus

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unix 3 [] STREAM CONNECTED 9931 /run/systemd/journal/socket

unix 3 [] DGRAM CONNECTED 8939 /run/systemd/journal/socket

unix 6 [] DGRAM CONNECTED 4663 /run/systemd/journal/socket

unix 3 [] STREAM CONNECTED 9417 /run/systemd/journal/stdout

unix 3 [] STREAM CONNECTED 9639 /run/systemd/journal/stdout

unix 3 [] STREAM CONNECTED 8880 /run/systemd/journal/stdout

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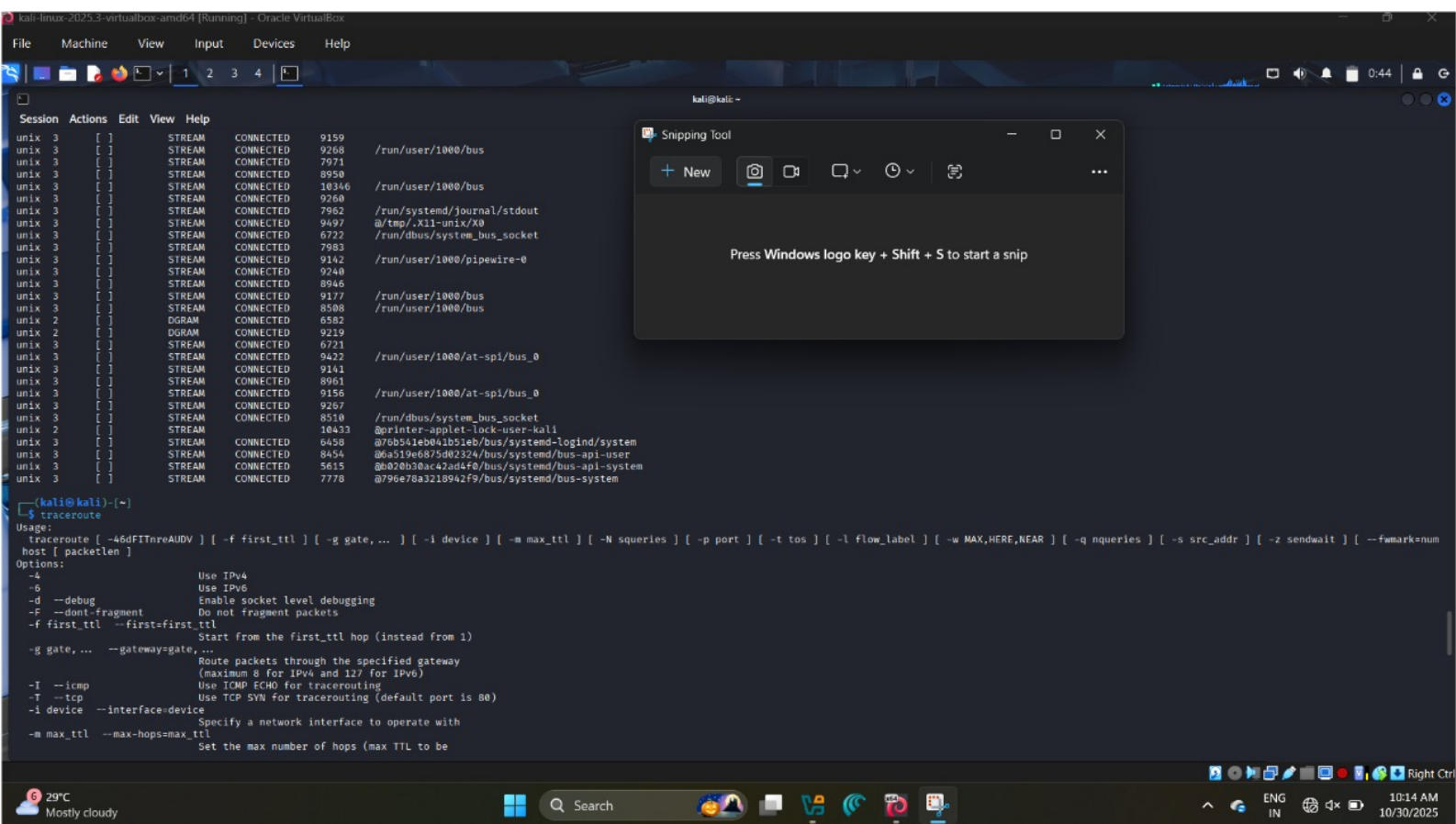
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Step 4: Networking Basics

Objective:

To understand how devices communicate over a network and the functions of key network layers and protocols used in cybersecurity.

Theory:

Networking is the process of connecting two or more computing devices to share data, resources, and services. In cybersecurity, understanding networking is crucial because all cyber-attacks and defenses occur through network communication.

The OSI Model (Open Systems Interconnection) divides communication into 7 layers:

1. Physical Layer – Deals with hardware like cables, switches, and signals.
2. Data Link Layer – Handles MAC addressing and error detection.
3. Network Layer – Uses IP addresses and routes data packets (e.g., routers).
4. Transport Layer – Ensures reliable transmission (TCP/UDP).
5. Session Layer – Manages sessions between devices.

6. Presentation Layer – Handles encryption, compression, and formatting.

7. Application Layer – Provides network services to users (HTTP, DNS, SMTP, etc.).

The TCP/IP model simplifies this into 4 layers: Application, Transport, Internet, and Network Access.

Common network commands in Linux (used in Kali):

ifconfig – Displays network interface details.

ping – Tests network connectivity.

netstat – Shows active ports and network connections.

traceroute – Displays the path packets take to reach a host.

nslookup – Resolves domain names to IP addresses.

Conclusion:

Networking forms the base for cybersecurity. Knowing how data travels helps in analyzing threats, packet captures, and attacks like sniffing or spoofing.

Step 5: Cryptography Basics

Objective:

To understand how cryptography protects data confidentiality, integrity, and authenticity.

Theory:

Cryptography is the science of securing communication by converting information into an unreadable format for unauthorized users.

Types of Cryptography:

1. Symmetric Encryption:

Uses one key for both encryption and decryption.

Example Algorithms: AES, DES, 3DES.

Example Command:

```
openssl enc -aes-256-cbc -salt -in file.txt -out file.enc
```

2. Asymmetric Encryption:

Uses two keys – a public key (for encryption) and a private key (for decryption).

Example Algorithm: RSA.

Example Command:

```
openssl genpkey -algorithm RSA -out private.pem
```

```
openssl rsa -pubout -in private.pem -out public.pem
```

3. Hashing:

Converts data into a fixed-length code, used to verify data integrity.

Example Algorithms: MD5, SHA-1, SHA-256.

Example Command:

```
md5sum file.txt
```

```
sha256sum file.txt
```

4. Digital Signatures:

Prove authenticity and integrity of a message using private/public key pairs.

Example Command:

```
openssl dgst -sha256 -sign private.pem -out sign.bin file.txt
```

```
openssl dgst -sha256 -verify public.pem -signature sign.bin file.txt
```

Conclusion:

Cryptography ensures the security of sensitive information during storage or transmission. It is the backbone of secure communications like HTTPS, digital banking, and authentication systems.



Step 6: Tool Familiarization

Objective:

To explore and understand the working of basic cybersecurity tools used in Kali Linux for analysis, scanning, and testing.

Theory:

Kali Linux provides a wide range of tools for ethical hacking and penetration testing. Understanding these tools is essential to perform network and system security analysis.

1. Wireshark (Packet Capture Tool)

Wireshark is a network protocol analyzer that captures and inspects packets in real time.

It helps identify suspicious traffic, monitor connections, and troubleshoot networks.

Command:

```
sudo wireshark &
```

Use case: Capture packets while pinging a website or browsing.

2. Nmap (Network Scanner)

Nmap (Network Mapper) is a tool for discovering hosts and services on a network.

It can perform port scanning, OS detection, and vulnerability scanning.

Command:

```
nmap -sP 192.168.1.0/24
```


Use case: Identify active devices and open ports.

3. Burp Suite (Web Proxy Tool)

Burp Suite is used for web application security testing.

It intercepts HTTP/S requests and allows modification before sending to the server.

Steps: Start Burp → Configure proxy (127.0.0.1:8080) → Intercept browser traffic.

4. Netcat (Network Debugging Tool)

Netcat (nc) is used to create network connections for debugging and data transfer.

Acts as a simple chat or file transfer tool between two systems.

Commands:

On one terminal: `nc -lvp 4444`

On another: `nc 127.0.0.1 4444`

Conclusion:

These tools are essential for ethical hackers to monitor, analyze, and test system vulnerabilities in a controlled environment.